

Confusion Buffer and Anki Starter

The Confusion Buffer — One-Page Reference + Anki Starter Deck

Your shock absorber for confusing terminology. Print this. Keep it next to your desk.

The 8 rules (condensed)

1. **Intuition first, always.** Visual explainer (3B1B / StatQuest / a blog) → **then** the rigorous source → **then** derive/implement by hand. Never open a rigorous lecture cold.
2. **Living glossary.** One doc. Every confusing term → a plain-English line + a tiny example + where it shows up. Add nightly. You look terms up in *your own words*, not from scratch.
3. **Spaced re-derivation (Anki).** Anything hard → a card. Review at **1 d → 3 d → 1 wk → 1 mo**. Concepts get muscle memory, just like code.
4. **The Feynman gate = "done."** Not done until you can explain it out loud, plain language, **to a 12-year-old, no notes**. Can't? You just found the exact gap — go back to rule 1 for that piece.
5. **Unblock ritual (no grinding).** Stuck >30 min → write the exact confusion in a log, then **switch teacher/modality**. A different angle usually unlocks it fast.
6. **Consolidation weeks (~every 5th).** No new material — only re-derive, review the glossary, re-implement the shakiest thing. This is what stops confusion compounding.
7. **Gate on understanding, not the calendar.** Self-tests are pass/fail on *explain + derive + implement unaided*. Don't advance until you pass — let the timeline stretch.
8. **Triangulate the hard six.** Backprop · Attention/ $\sqrt{d_k}$ · SVD/eigen · Diffusion · Equivariance · VQE/barren-plateaus each have 2-3 teachers listed in the plan. If one doesn't click, the next will.

Daily ritual: thread (45 min) → phase work → 15 min glossary + Anki. **Weekly:** one Whiteboard-Friday concept (photo it), one Sunday teach-a-junior writeup.

Anki starter deck — the hard six (copy these in as your seed cards)

Card format is Q (front) → A (back). Make a deck per topic; add your own as you go. Prefer "explain/derive" cards over "recall a fact" cards — they train the interview skill.

1 • Backpropagation

- **Q:** In one sentence, what is backprop doing? → **A:** Applying the chain rule to multiply local derivatives along every path from a parameter to the loss, so each parameter learns how much it contributed to the error.
- **Q:** For an MLP layer $a_l = f(W_l a_{l-1} + b_l)$, what is dL/dW_l ? → **A:** $dL/dW_l = \delta_l \cdot a_{l-1}^T$, where $\delta_l = (dL/dz_l)$ is the layer's error signal; the outer product has the same shape as W_l .
- **Q:** How does the error signal δ propagate backward one layer? → **A:** $\delta_{l-1} = (W_l^T \delta_l) \odot f'(z_{l-1})$ — pull the error back through the weights, then gate by the local activation derivative.
- **Q:** Why do vanishing gradients happen and what fixes them? → **A:** Repeated multiplication by small f' (e.g. sigmoid saturating) shrinks gradients exponentially with depth; fixes: ReLU-family activations, residual connections, normalization, careful init.

2 • Attention & the $\sqrt{d_k}$ scaling

- **Q:** Write scaled dot-product attention. → **A:** $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$, where d_k is the key dimension.
- **Q:** Why divide by $\sqrt{d_k}$? → **A:** With unit-variance Q, K entries, the dot product of two d_k -vectors has variance $\sim d_k$; dividing by $\sqrt{d_k}$ rescales logits to unit variance so softmax doesn't saturate into near-one-hot (which would kill gradients).
- **Q:** What do Q, K, V represent intuitively? → **A:** Query = "what am I looking for", Key = "what do I offer", Value = "what I pass on if matched"; attention weights are query-key similarities, output is a weighted sum of values.
- **Q:** Why multi-head instead of one big head? → **A:** Multiple lower-dim heads let the model attend to different relationships/subspaces in parallel, then concatenate — more expressive than a single softmax over one large space.

3 • SVD & eigendecomposition

- **Q:** State the SVD of a real matrix A ($m \times n$). → **A:** $A = U \Sigma V^T$: U ($m \times m$) and V ($n \times n$) orthogonal, Σ diagonal with non-negative singular values; columns of V are right-singular vectors, of U left-singular vectors.
- **Q:** How is SVD related to eigendecomposition? → **A:** Right-singular vectors of A = eigenvectors of $A^T A$; singular values = $\sqrt{\text{eigenvalues of } A^T A}$; left-singular vectors = eigenvectors of AA^T .
- **Q:** Why is $A^T A$ always positive semi-definite? → **A:** For any x , $x^T A^T A x = \|Ax\|^2 \geq 0$; so all eigenvalues are ≥ 0 .
- **Q:** What is the best rank- r approximation of A (and which theorem)? → **A:** Keep the top- r singular values/vectors: $A_r = \sum_{i \leq r} \sigma_i u_i v_i^T$ (Eckart-Young); minimizes $\|A - A_r\|$ in Frobenius/spectral norm.

4 • Diffusion / score-matching

- **Q:** What are the forward and reverse processes in a DDPM? → **A:** Forward: gradually add Gaussian noise over T steps until $\sim$ pure noise. Reverse: a learned model denoises step-by-step from noise back to a sample.
- **Q:** What does the network actually predict, and what's the loss? → **A:** Typically the added noise ϵ ; the simple loss is $\mathbb{E} \|\epsilon - \epsilon_\theta(x_t, t)\|^2$ — a re-weighted variational bound (ELBO).
- **Q:** What is the "score" and how does score-matching connect to diffusion? → **A:** The score is $\nabla_x \log p(x)$; denoising is equivalent to estimating the score of the noised data, so diffusion = learning scores across noise levels, then following them to sample.
- **Q:** Two views of the diffusion objective? → **A:** (i) Variational: maximize an ELBO on data likelihood; (ii) Score-matching: match $\nabla_x \log p$ at each noise level — both yield the same denoising loss.

5 • Equivariance (geometric / molecular ML)

- **Q:** Define invariance vs equivariance. → **A:** f is **invariant** if $f(g \cdot x) = f(x)$; **equivariant** if $f(g \cdot x) = g \cdot f(x)$ — the output transforms the same way as the input under group action g .
- **Q:** Why does E(3)-equivariance matter for molecules? → **A:** Physical properties are invariant/equivariant to rotation, translation, reflection; baking that in means the model doesn't waste capacity/data relearning symmetry, and predictions are consistent under pose changes.
- **Q:** How does a message-passing GNN update a node? → **A:** For each node: aggregate (permutation-invariantly, e.g. sum) messages from neighbors computed from neighbor+edge features, then update the node state — repeated over layers.
- **Q:** Where does AlphaFold use equivariance? → **A:** Its structure module operates on residue frames so that predicted 3D coordinates transform correctly (equivariantly) under global rotation/translation.

6 • VQE & barren plateaus

- **Q:** What does VQE minimize, and on what principle? → **A:** It minimizes $\langle \psi(\theta) | H | \psi(\theta) \rangle$ over circuit parameters θ (the variational principle: this expectation $\geq$ the true ground-state energy), to estimate a molecule's ground-state energy.
- **Q:** How are gradients taken on a quantum circuit? → **A:** The **parameter-shift rule**: $\partial \langle H \rangle / \partial \theta = \frac{1}{2} [\langle H \rangle(\theta + \pi/2) - \langle H \rangle(\theta - \pi/2)]$ — exact gradients from evaluating the circuit at shifted parameters.
- **Q:** What is a barren plateau? → **A:** For many random/expressive parameterized circuits, gradient variance vanishes exponentially in qubit count, so the cost landscape is almost flat — training stalls. Mitigations: structured ansätze, local cost functions, smart initialization.
- **Q:** VQE vs QAOA — one line each. → **A:** VQE: general variational eigensolver for Hamiltonian ground states (e.g. chemistry). QAOA: a specific alternating-operator ansatz aimed at combinatorial optimization.

Add cards whenever the Feynman gate exposes a gap. A term you had to look up twice is a term that belongs in Anki. Review daily; the 1→3→7→30-day spacing does the rest.